

TECHNICAL REPORT

Road Network Extraction

from High-Resolution Satellite Imagery

Using Domain-Adaptive Deep Learning

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Abstract

This report details the development, training, and evaluation of a deep learning system for automated road network extraction from high-resolution satellite imagery. The methodology was structured in two sequential stages to address the problem of domain shift. The first stage established a global baseline using the DeepGlobe dataset. The second stage applied a domain adaptation strategy through multi-city fine-tuning on high-resolution Indian urban imagery.

The final architecture incorporates Test-Time Augmentation (TTA) and morphological post-processing, producing significant improvements in structural connectivity and segmentation accuracy on previously unseen terrain. The model achieves an IoU of 65.69% and an F1 score of 79.29% on the held-out Indore test region under zero-shot inference conditions.

65.69%	79.29%	85.37%	74.02%
IoU	F1 Score	Recall	Precision

1. Objective

The primary objective of this project was to design, train, and validate a deep learning model capable of extracting road networks from satellite imagery with high metric fidelity. Road extraction is a foundational task in remote sensing, with downstream applications in urban planning, autonomous navigation, and

geographic information system (GIS) automation. Traditional manual digitization is labor-intensive and does not scale to the acquisition rates of modern Earth observation satellites.

This study addresses the scaling challenge through a two-stage approach: training a baseline segmentation model on globally diverse data, then applying targeted regional fine-tuning to overcome geographical domain shift. A key requirement was that the final model be evaluated purely on zero-shot inference against a geographically isolated urban region.

2. Approach and Methodology

2.1 Model Architecture

The segmentation architecture is a U-Net with a ResNet-50 encoder, chosen for its proven effectiveness in dense binary segmentation tasks.

- **Encoder (ResNet-50):** Pre-trained on ImageNet-1K, providing strong initialisation of general visual features and reducing the volume of task-specific data required.
- **Decoder with Skip Connections:** Sequentially up-samples feature maps back to input resolution. Skip connections at each hierarchical level concatenate high-resolution encoder features into the decoder, preserving fine geometric boundaries critical for thin road delineation.

We also tried different model architectures like SegFormer-B3 but it resulted in around same evaluation metrics.

2.2 Loss Formulation

Training used a composite loss combining Binary Cross-Entropy (BCE) and Soft Dice loss. BCE included a positive-class weighting factor to penalise false negatives, directly addressing the severe foreground-background class imbalance where road pixels constitute a small fraction of the image. Soft Dice loss reinforced spatial overlap maximisation and stabilised optimisation.

2.3 Training Strategy

- **Baseline Training:** Conducted over 60 epochs using AdamW optimisation on the DeepGlobe dataset, establishing a generalised global baseline without augmentation overhead.
- **Fine-Tuning:** Fine-tuning on five Indian cities using Cosine Annealing learning rate scheduling, with heavy sampling from high road-density regions to mitigate domain shift.

2.4 Model Hyperparameters & Configurations

Category	Parameter	Value
Architecture	Base Model	U-Net
Architecture	Encoder	ResNet-50 (ImageNet Pre-trained)
Architecture	Patch Size	512 x 512
Training	Optimizer	AdamW
Training	Learning Rate	1e-4 (Initial)
Training	Weight Decay	1e-3
Training	Scheduler	Cosine Annealing (T_max=30)
Training	Batch Size	8
Training	Epochs	60 (Baseline), 30 (Fine-Tuning)
Training	Loss Function	0.5 x BCE + 0.5 x Soft Dice

3. Data

Dataset usage was strategically separated across both development stages to ensure robust domain alignment and clean evaluation protocols.

- **DeepGlobe Benchmark (Baseline):** Approximately 10,500 annotated, high-resolution RGB patches representing global road topology. Used exclusively for baseline training.
- **Indian Urban Dataset (Fine-Tuning):** High-resolution imagery from Bhopal, Jabalpur, Gwalior, Ujjain, and Nagpur. Used for domain-specific fine-tuning.
- **Held-Out Test Set (Indore):** Imagery from Indore, strictly isolated from all training and fine-tuning pipelines. Used solely for the final quantitative evaluation to ensure a genuine zero-shot benchmark.

4. Image Processing and Post-Processing

Multiple processing pipelines were implemented to handle the structural and photometric variability present in satellite data.

- **Data Augmentation:** The training dataset was augmented using geometric transformations (orthogonal rotations, elastic deformations) and photometric adjustments (hue-saturation shifts) to improve model robustness.
- **Test-Time Augmentation (TTA):** During inference, multiple spatial permutations (rotations and flips) of each input patch were independently evaluated. The resulting probability maps were averaged to produce a stabilised final prediction.

5. Challenges

- **Thin Road Detection:** Rural roads and unpaved tracks frequently approached the physical resolution limit of 0.5m/px satellite imagery, causing extraction fade-out and structural fragmentation.
- **False Positives in Urban Areas:** Flat concrete rooftops, dry canal systems, and pedestrian walkways share spectral and geometric properties with roads, occasionally generating false-positive classifications in dense urban areas.
- **Domain Shift:** The DeepGlobe baseline was trained on globally diverse imagery with different urban layout priors. Indian cities, with their dense clustering and irregular road geometry, caused immediate inference degradation before fine-tuning was applied.
- **Computational Constraints:** High-resolution feature extraction imposed substantial GPU memory demands, restricting batch sizes and limiting iteration speed during baseline training.

6. Experimental Results

6.1 Baseline Performance and Domain Shift

The baseline model was evaluated on its native DeepGlobe test split and then on the unseen Indian topology (Indore) to measure domain generalisation.

Training Time – around 30 hours.

Evaluation Set	IoU	F1 Score	Notes
DeepGlobe (native test split)	66.66%	0.7847	<i>In-domain</i>
Indore — before fine-tuning	42.12%	0.5927	<i>Zero-shot</i>

The 24.5-point IoU drop when transferring directly from DeepGlobe to Indore confirms the presence of significant domain shift and motivates the targeted fine-tuning approach.

6.2 Fine-Tuned Model Performance (Model)

After fine-tuning, Model was evaluated against the strictly held-out Indore dataset with TTA and morphological post-processing applied.

Training Time – around 8 hours.

Metric	IoU	F1 Score	Precision	Recall
Model (Indore)	65.69%	79.29%	74.02%	85.37%

Pixel-level confusion matrix on the Indore test set:

Category	Predicted Road	Predicted Background
Actual Road	TP: 18,769,556	FN: 3,216,924
Actual Background	FP: 6,586,262	TN: 84,451,680

Model v3 recovered IoU from 42.12% (pre-fine-tuning) to 65.69%, closely approaching the 66.66% achieved on the native DeepGlobe test split. The high recall of 85.37% confirms that the model correctly captures the vast majority of road pixels, while the precision of 74.02% reflects the accepted trade-off with morphological closing in dense urban areas.

7. Visual Inference Outputs

Qualitative evaluation of the segmentation model requires visual inspection of inference outputs. The figure below shows a representative sample from the Indore test set.



Figure 1: Left — Raw satellite input; Centre — Model v3 predicted road overlay; Right — Ground truth overlay.

- **Input Image:** The raw high-resolution RGB satellite patch as processed by the network encoder.
- **Ground Truth Mask:** The pixel-level binary annotation delineating the precise road extents as manually traced.
- **Predicted Mask:** The binary segmentation map produced by Model v3 after morphological post-processing.
- **Overlay Result:** A composite overlay of the predicted mask on the raw input, enabling qualitative assessment of boundary alignment and false-positive patterns.

8. Conclusion

This report documents the full development of a road extraction pipeline, from global baseline training on DeepGlobe to targeted regional fine-tuning on Indian urban datasets. The quantitative results validate the importance of domain adaptation: fine-tuning recovered IoU from 42.12% to 65.69% on a held-out city never seen during training.

The final Model demonstrates that combining a well-established segmentation backbone (U-Net + ResNet-50) with precision-focused post-processing techniques produces reliable road network extractions even under challenging zero-shot conditions. A hyperparameter summary will be appended separately.

9. Team Members and Contributions

Team Member	Role	Key Contributions
Vikash Kumar Singh	Dataset Preparation & Baseline Training	Satellite imagery procurement and GeoJSON preprocessing; dataset augmentation pipeline; baseline U-Net training and initial evaluation on DeepGlobe.
Ayush Kumar	Fine-Tuning & Inference Pipeline	Multi-city data loading and sampling strategy; TTA implementation and morphological post-processing framework; final zero-shot evaluation on the Indore test set.

References

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